ITAI-1371 - Intro to Machine Learning

Reflective Journal

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In Lab 06, most of the machine learning workflow was already provided in the Google Colab notebook. My work was not to build the full program from scratch. Instead, I followed the provided code, corrected the line breaks in the regression data preparation section, and ran it. I also added a verification cell that showed zero missing age values, 571 training samples, and 143 testing samples. This helped me see how the code prepared and split the data before the model used it.

Even though the notebook was mostly prepared, working through it helped me understand the purpose of the steps. The code loaded the Titanic dataset, removed rows with missing ages, selected features and a target, and split the data into training and testing sets. I learned that running a cell is not enough to know whether the result makes sense. Checking the missing values and sample counts helped confirm what the code had done.

I also answered Knowledge Check Question 1, which asked about regression and classification. I learned that regression predicts a numerical value, such as a fare, while classification predicts a category, such as survived or did not survive. Both can be supervised learning because the notebook provides a known target for the model to learn from. The type of target also affects how results are evaluated: regression uses measures such as mean squared error, while classification can use accuracy.

This lab helped me connect data preparation to the rest of the machine learning process. The code for training and evaluating the models was already in the notebook, but I could follow how those later steps depended on preparing the data first. I also wondered whether removing rows with missing ages is always the best choice, or whether filling in the missing values would be more appropriate in another dataset. That would depend on the data and the goal of the analysis.

Danh Nguyen

Before this lab, I treated “accuracy” as a general score for any model, but answering the Knowledge Check questions changed how I think about this. I learned that regression and classification are really two different kinds of questions. Regression asks “how much?” and classification asks “which one?”, and that difference decides which model and metric to use. The idea that clicked for me was imagining the model predicting a $30.10 fare when the real fare was $30.00. Accuracy would count that as wrong, the same as predicting $300, which is why MSE fits regression better. Interpreting lr_model.coef_ also showed me that you have to understand the data to read a model correctly. The negative Pclass coefficient only made sense once I remembered that a higher class number means a cheaper ticket.

Great Inioluwa Olayiwola

1. Understanding the ML Workflow

Working through the machine learning workflow revealed how each step logically builds toward evaluating model performance rather than just being a sequence of arbitrary tasks. The most critical insight was understanding why we split data into training (80%) and testing (20%) sets: the training set allows the model to learn underlying patterns, while the test set acts as an unbiased evaluation on unseen data to ensure the model isn't simply memorizing inputs. Additionally, preparing data such as encoding the categorical 'Sex' column into binary numerical values (0 and 1) highlighted that machine learning algorithms are fundamentally mathematical engines that require clean, numeric input to function properly

2. Regression vs. Classification

The lab clearly demonstrated the fundamental distinction between regression and classification problems based on the nature of their target variables. Regression tasks, such as predicting a passenger's continuous ticket 'Fare', require metrics like Mean Squared Error (MSE) to quantify how far off predictions are from actual values on a continuous scale. In contrast, classification tasks, like predicting categorical 'Survived' outcomes (0 or 1), rely on Accuracy Score to measure the proportion of correct discrete guesses. Applying accuracy to a regression problem like fare prediction would be ineffective because hitting an exact dollar amount down to the penny is extremely rare, making continuous error measurement essential for evaluating numeric predictions.

3. Demystifying Model Weights (.coef_)

Inspecting the .coef_ attribute of the trained Linear Regression model demystified how the algorithm transforms input features into final predictions. These coefficients represent the mathematical weights assigned to each feature, indicating both the strength and direction of their influence on the target variable. For instance, a negative coefficient assigned to passenger class (Pclass) aligns with domain intuition, as moving to a higher numerical class (e.g. 3rd class) corresponds to lower ticket fares. Understanding these weights converts the model from a "black box" into an interpretable system where individual feature contributions can be measured and validated.

4. Key Takeaways & Next Steps

This exercise reinforced that model metrics must be translated into real-world units to be meaningful, as seen when taking the square root of the MSE (approximately 58.01) to realize predictions were off by about 58 on average. Moving forward, improving model performance will likely require incorporating richer feature sets beyond just age and class, alongside learning how to handle missing data without discarding valuable rows. Ultimately, this lab shifted my perspective from viewing machine learning as writing code to viewing it as a structured process of preparing data, selecting appropriate algorithms, and critically interpreting model outputs.

Karima Sadruddin

This task helped me understand that logistic regression is used for classification, such as predicting whether a Titanic passenger survived. The model learns patterns from the training data and uses Age, passenger class (Pclass), and Sex to make predictions. I learned that accuracy is the proportion of test-set predictions that match the actual survival labels.

However, accuracy does not show every type of mistake, and this model may miss important influences on survival because it uses only three features. Running the notebook and reviewing its output will help me confirm the result and explain it accurately. AI assistance disclosure: ChatGPT helped organize and explain the code and helped draft this reflection. I should review the wording and make sure it reflects my own understanding before submitting

Quan Thach

Working on this lab provided a practical look into the structural differences between regression and classification tasks using the Titanic dataset. Implementing the linear regression model to predict ticket fares highlighted how continuous numerical values are handled by calculating specific slopes for features like age and passenger class. Stepping through the code configuration required a clear understanding of data formatting, especially when ensuring that missing values were handled appropriately before feeding the feature arrays into the training pipeline.

Analyzing the resulting coefficients from the linear regression task offered valuable insights into feature importance and directionality. Seeing a large negative coefficient for the passenger class variable clearly demonstrated how moving down to a less luxurious tier corresponds to a sharp decline in predicted price. Conversely, the minimal coefficient for age showed how some features carry far less weight when holding other variables constant. This part of the exercise emphasized that machine learning is not just about generating a prediction, but also about interpreting what the model actually learns.

Transitioning to the logistic regression task underscored why different metrics are necessary for distinct machine learning problems. While mean squared error effectively measured the distance between predicted and actual numerical fares, it became clear that such a metric cannot apply to classification. Mapping variables like sex into binary indicators and using accuracy scores to evaluate survival predictions reinforced how classification focuses on group boundaries rather than numerical scales. Overall, completing these parallel tasks provided a balanced foundation for setting up, training, and properly evaluating baseline predictive models.